



Stellar structure and evolution

Pablo Marchant

Who is this guy?



Santiago, Chile
(Bachelor+Master)



PONTIFICIA
UNIVERSIDAD
CATÓLICA
DE CHILE

Who is this guy?



Bonn, Germany
(PhD)

Who is this guy?



Bonn, Germany
(PhD)



Who is this guy?



Chicago, USA
(PostDoc)

Who is this guy?



Chicago, USA
(PostDoc)



Who is this guy?



Leuven, Belgium

(PostDoc)

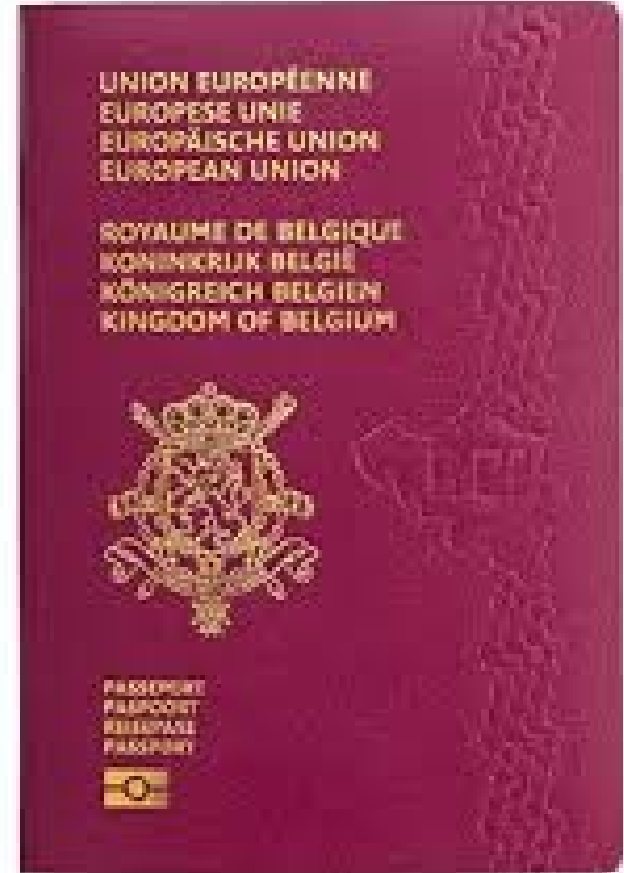
Who is this guy?



Leuven, Belgium

(PostDoc)

23rd September 2026







Big data stellar astrophysics

Big data stellar astrophysics



(Image credit: NSF-DOE Rubin Observatory/AURA/B. Quint)

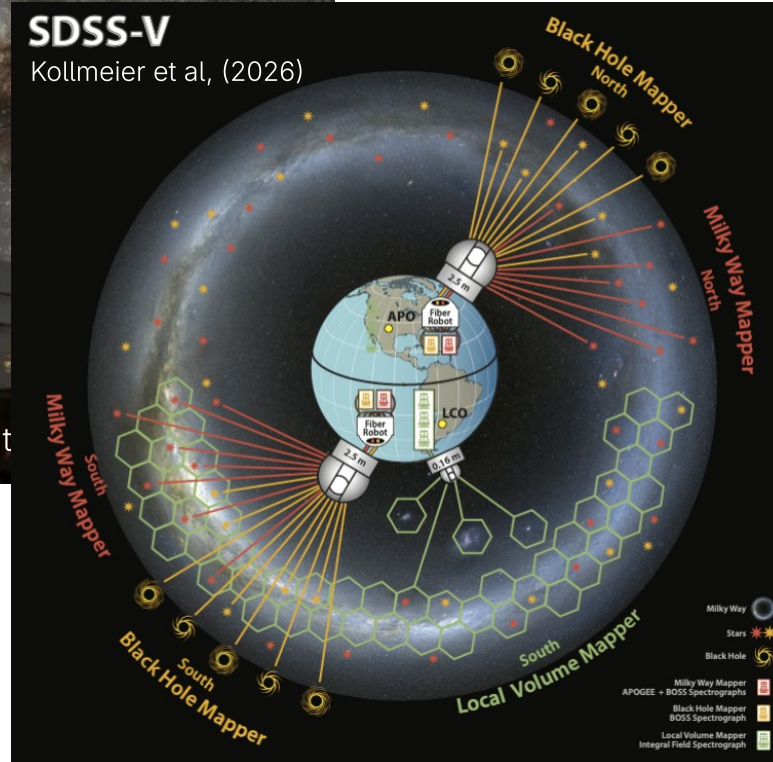
Ilezić et al. (2018)

Big data stellar astrophysics



(Image credit: NSF-DOE Rubin Observat

Ilezić et al. (2018)



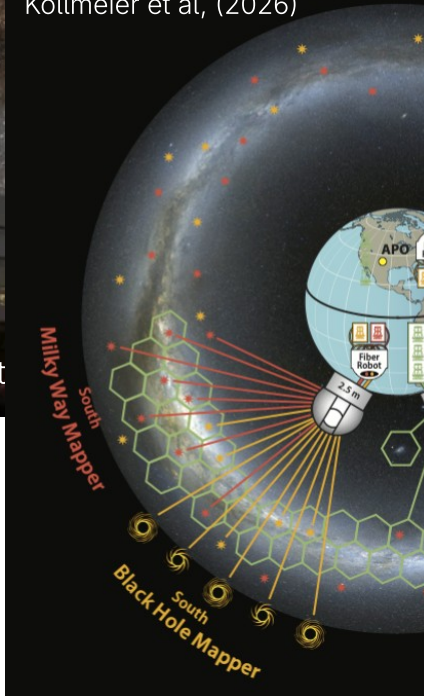
Big data stellar astrophysics



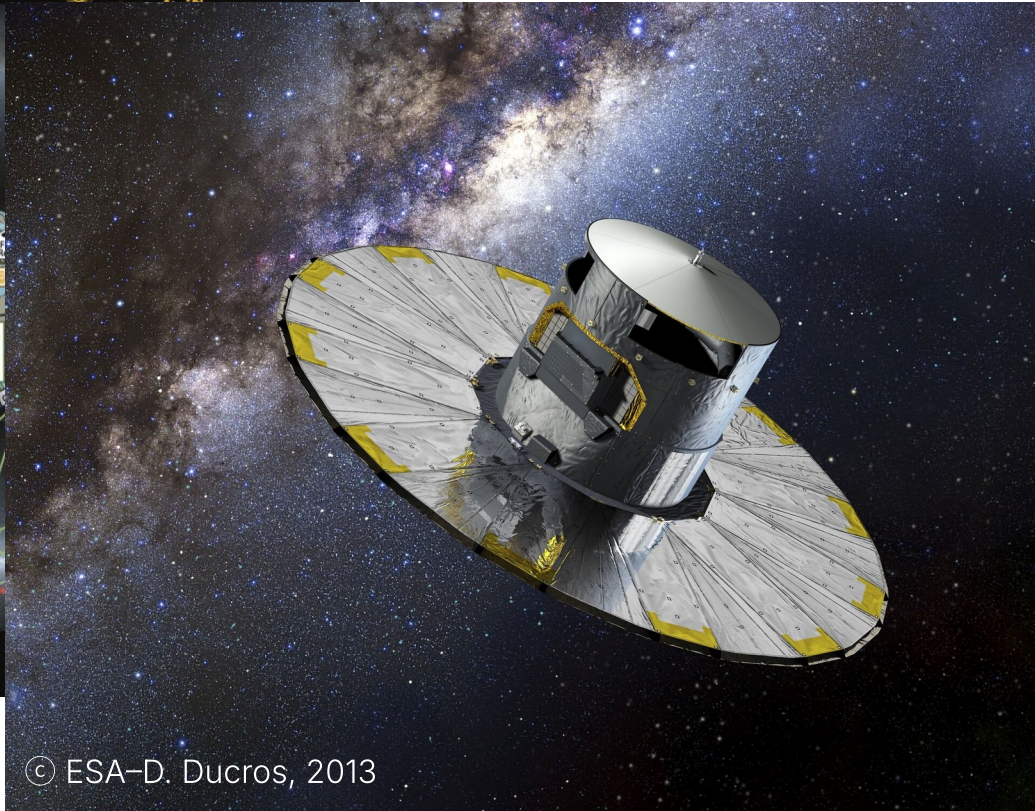
Ilezić et al. (2018)

SDSS-V

Kollmeier et al, (2026)

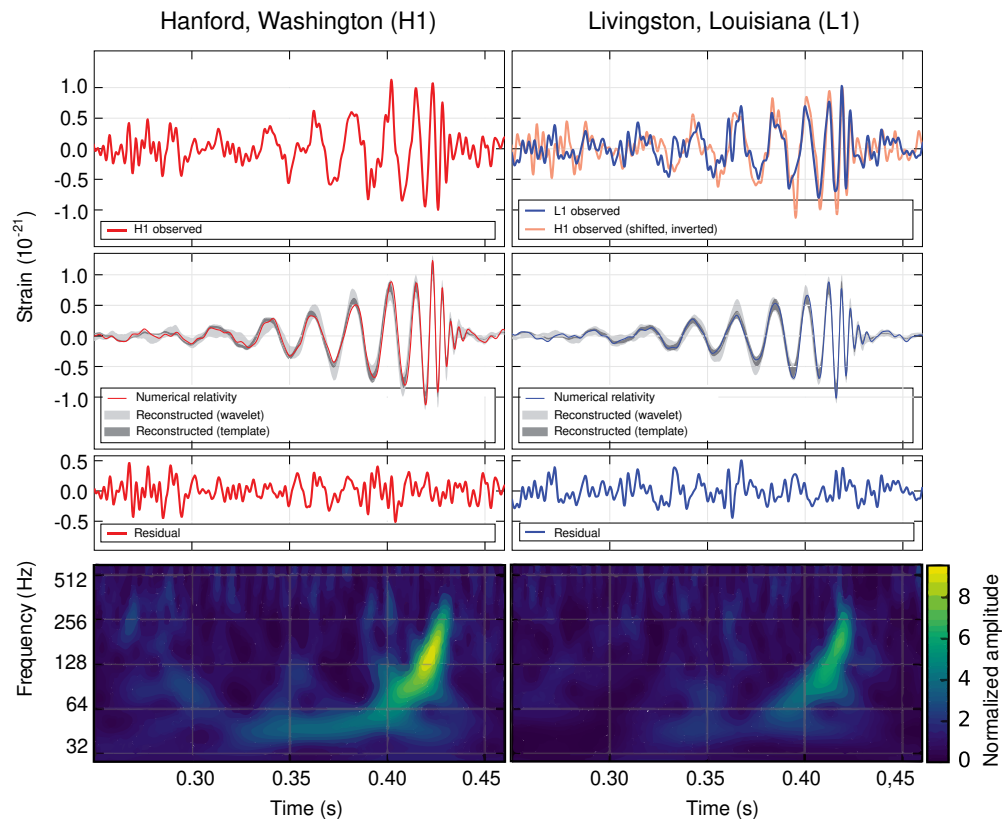


Gaia Collaboration (2016)



Extreme "laboratories"

When formed in close pairs, black holes can merge and produce detectable gravitational wave radiation.

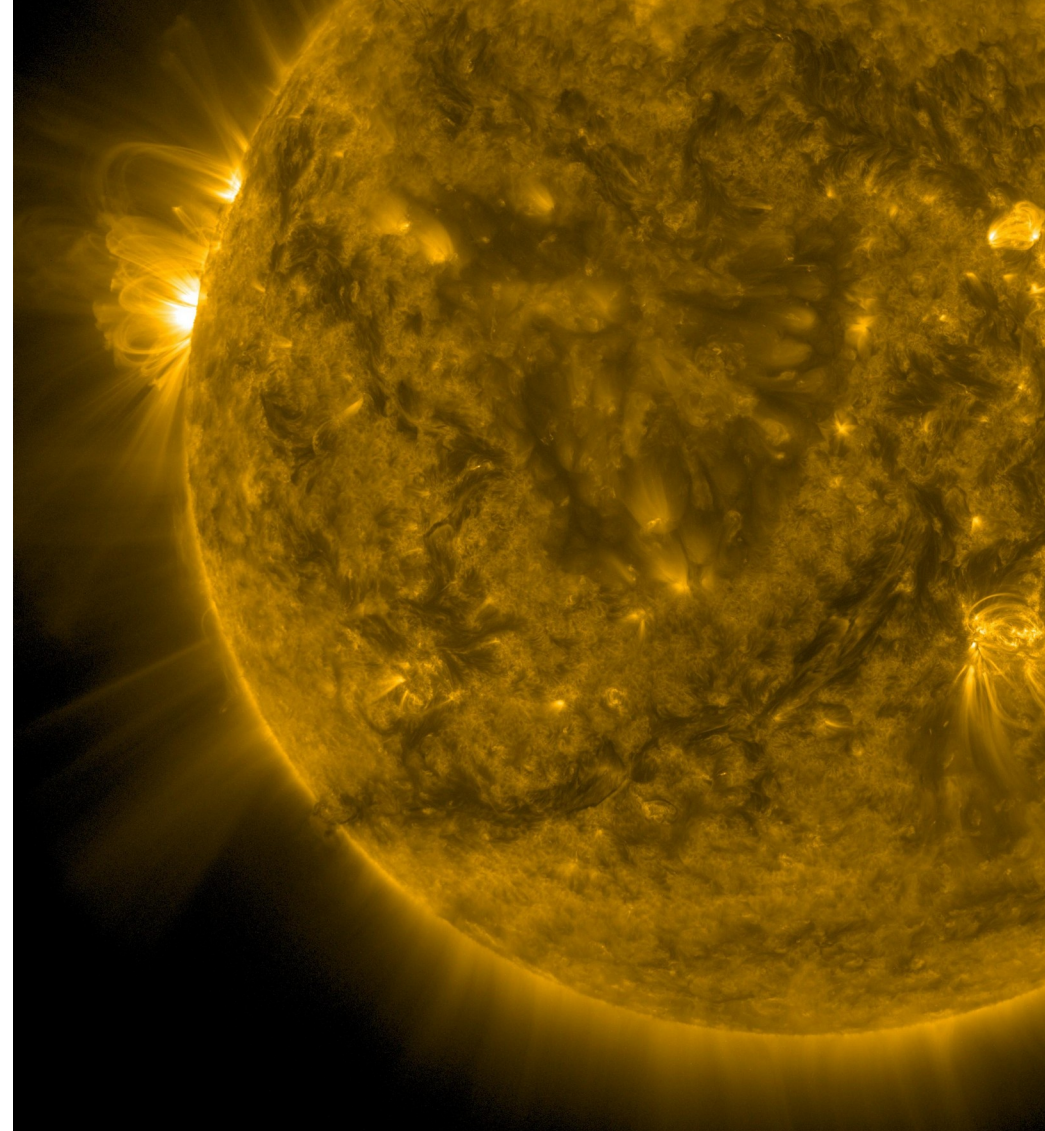


Abbott et al. (2016)



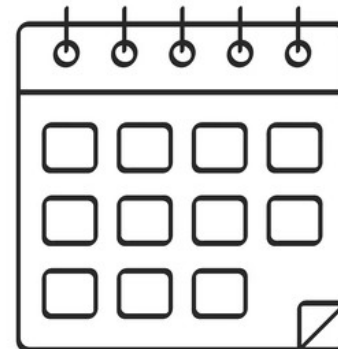
Objectives

- Derive the equations of stellar structure and evolution.
- Study the relevant microphysical ingredients (i.e., equations of state, nuclear reactions).
- Understand the different phases of evolution of a star, including its end stages.
- Compute detailed models of stellar evolution.



Course structure (tentative)

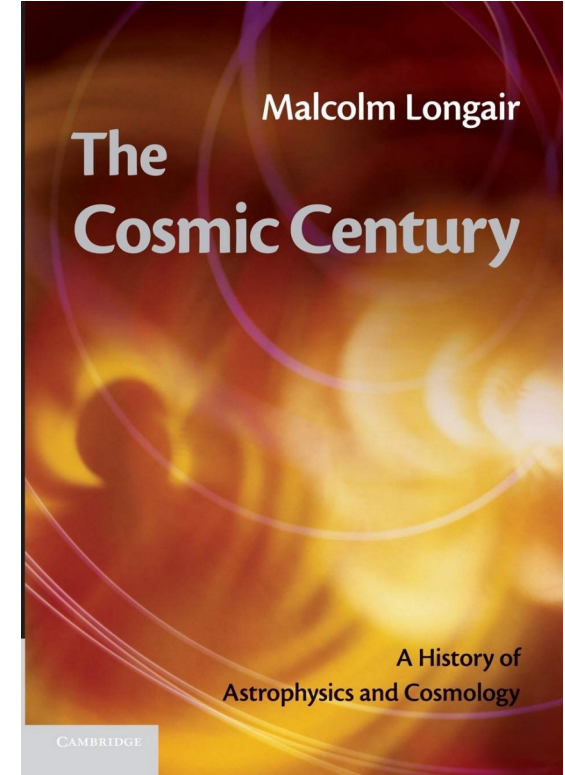
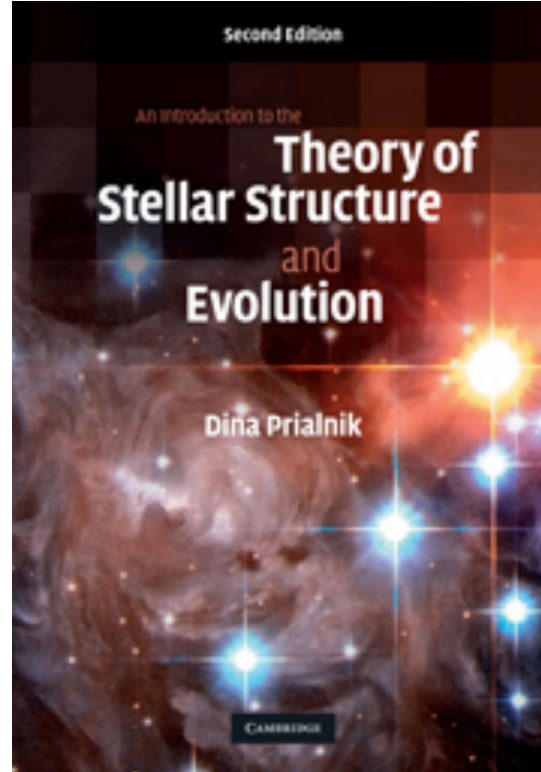
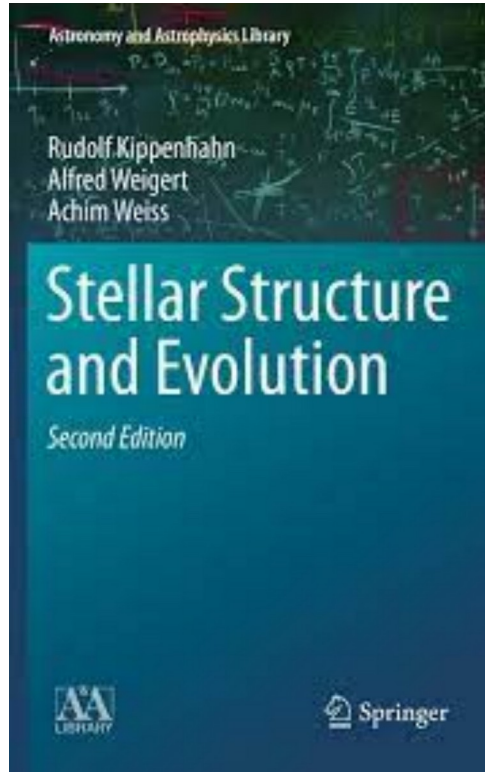
- **Today:** Intro + derivation of equations of stellar structure
- **30/09:** Equations of state I
- **07/10:** Equations of state II
- **14/10:** Radiative energy transfer
- **21/10:** Convection
- **28/10:** Nucleosynthesis
- **04/11:** Numerical methods
- **18/11:** Main-sequence evolution
- **25/11:** Post main-sequence evolution
- **02/12:** Binary stars
- **09/12:** Asteroseismology (or recap)



Evaluation:

- 8 points: "labs"
- 12 points: written exam

Resources



Measuring light

Apparent magnitude:

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{S_1}{S_2} \right)$$

Absolute magnitude: magnitude object would have at a distance of 10 parsec.

$$S = \frac{L}{4\pi d^2} \rightarrow M - m = 2.5 \log_{10} \left(\left[\frac{10 \text{ pc}}{d} \right]^2 \right)$$

Beware that this does not define the zero point!

Parallaxes

One parsec is the distance at which a star exhibits a parallax of one arcsecond.

$$\frac{d}{1 \text{ pc}} = \frac{1}{p/1''}$$

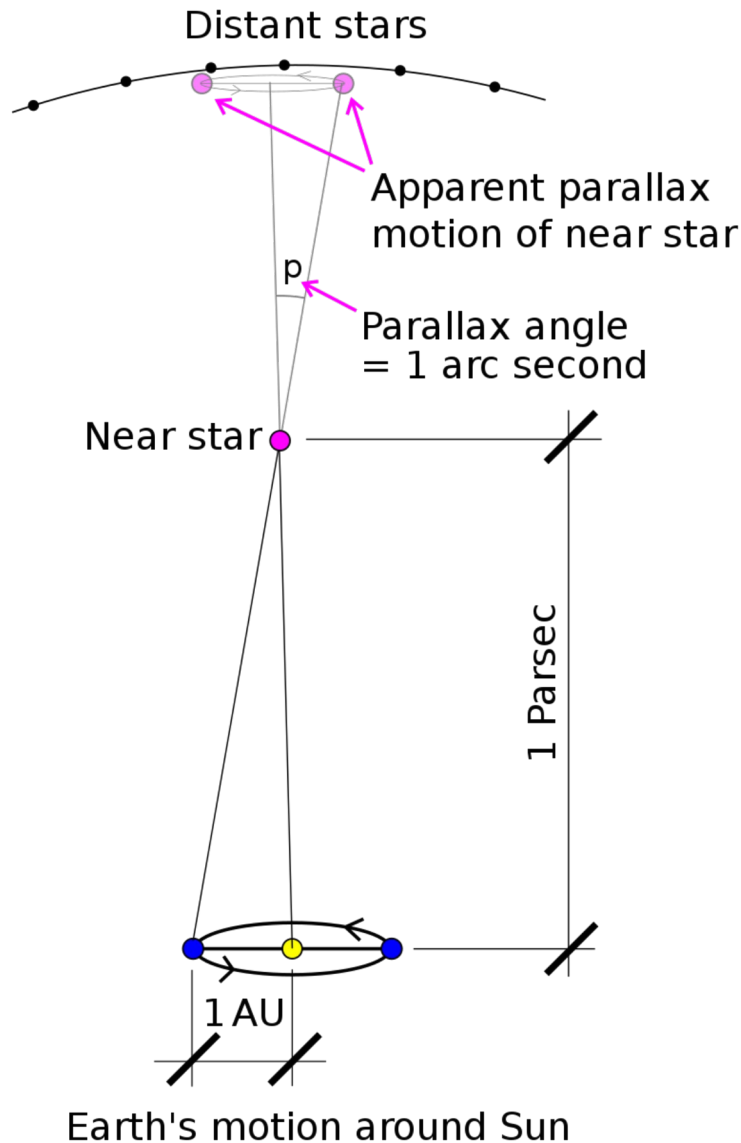
Astronomers use the CGS system!

$$1 \text{ pc} = 3.1 \times 10^{18} \text{ cm}$$

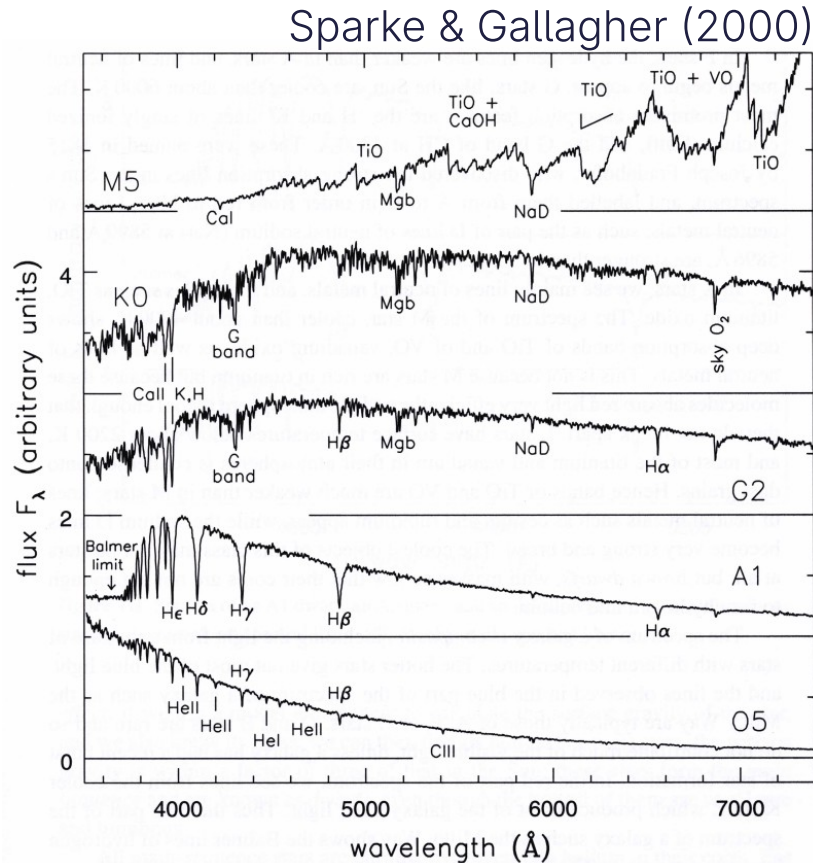
$$1 \text{ au} = 1.5 \times 10^{13} \text{ cm}$$

$$R_{\odot} = 7 \times 10^{10} \text{ cm}$$

$$R_{\oplus} = 6.4 \times 10^{10} \text{ cm}$$



Spectral energy distributions



To first order, the spectrum of most stars can be approximated as a black body with a specific effective temperature:

$$S_\lambda = \frac{R^2}{d^2} \frac{2hc^2\pi}{\lambda^5} \frac{1}{\exp(hc/\lambda k_B T) - 1}$$

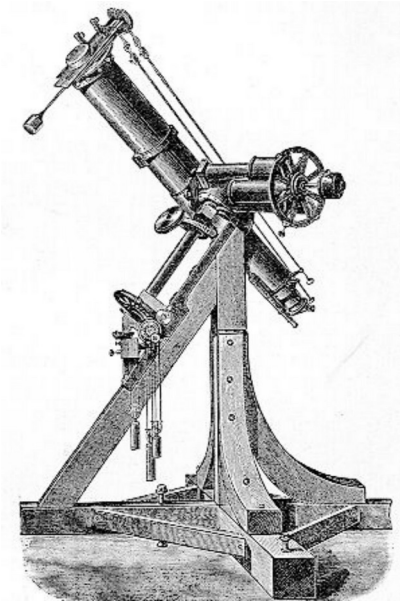
$$S = \int_0^\infty S_\lambda d\lambda = \frac{R^2}{d^2} \sigma T^4$$

In practice we only measure a fraction of this flux!

$$S_{\text{obs}} = \int_0^\infty S_\lambda \phi(\lambda) d\lambda$$

The initial drivers of astrophysics

Parallaxes



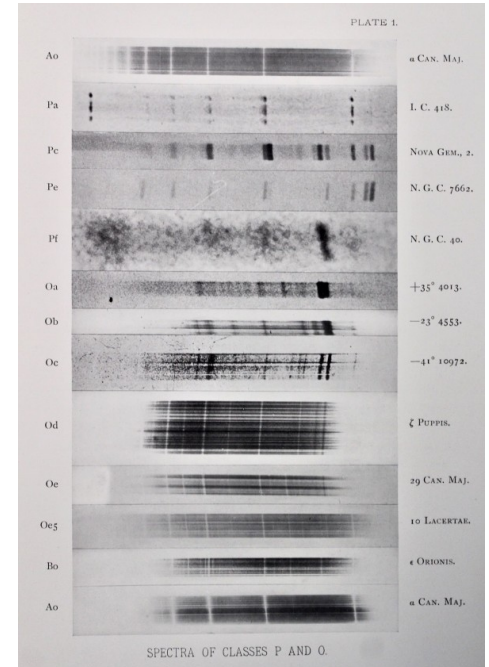
Bessel (1839),
Henderson (1840),
Struve (1840)

Photography



Great Orion Nebulae
by Henry Draper (1880)

Spectroscopy



Sample spectra
from A. Cannon

The Harvard computers



Draper memorial catalogue

(Pickering, 1890)

Including classification of 10351 stars

The Harvard computers

Williamina Fleming



Antonia Maury

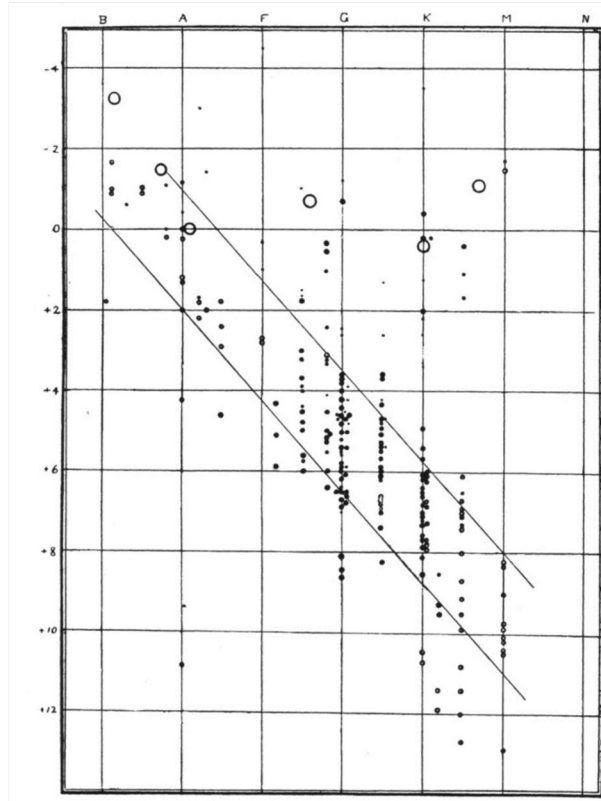


Annie Cannon

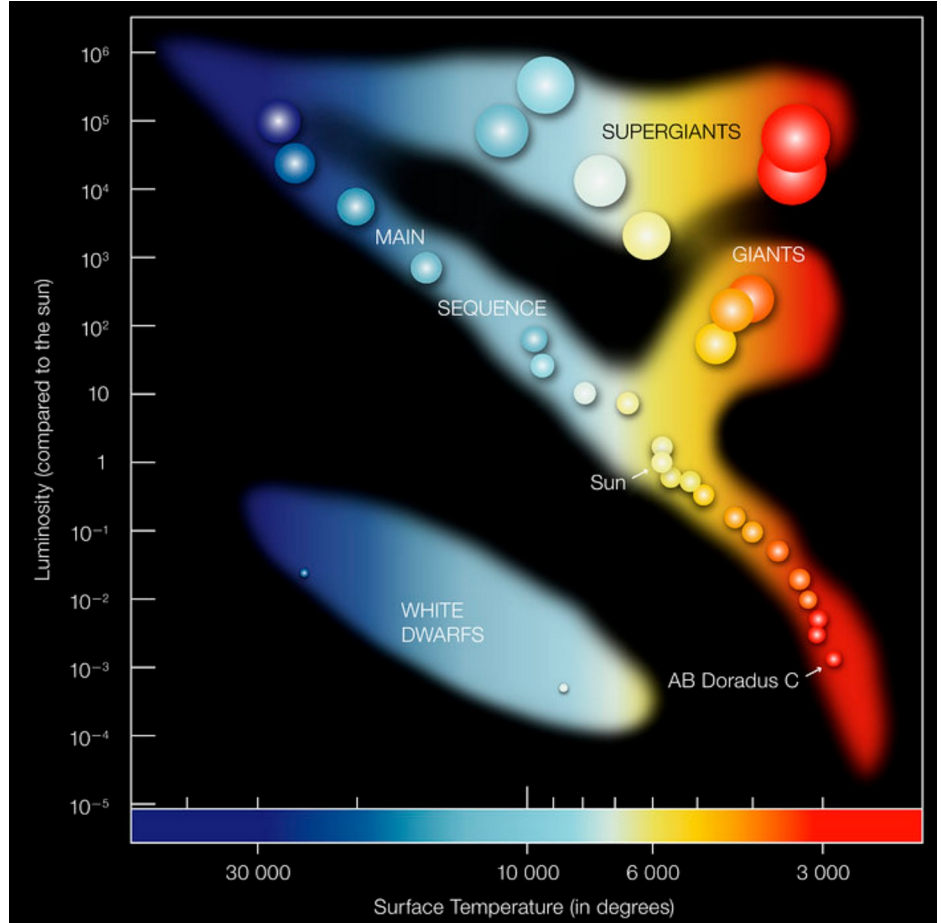


Developers of the **OBAFGKM** classification. Annie Cannon classified hundreds of thousands of spectra, forming the Henry Draper (HD) catalogue.

The Hertzsprung-Russell diagram

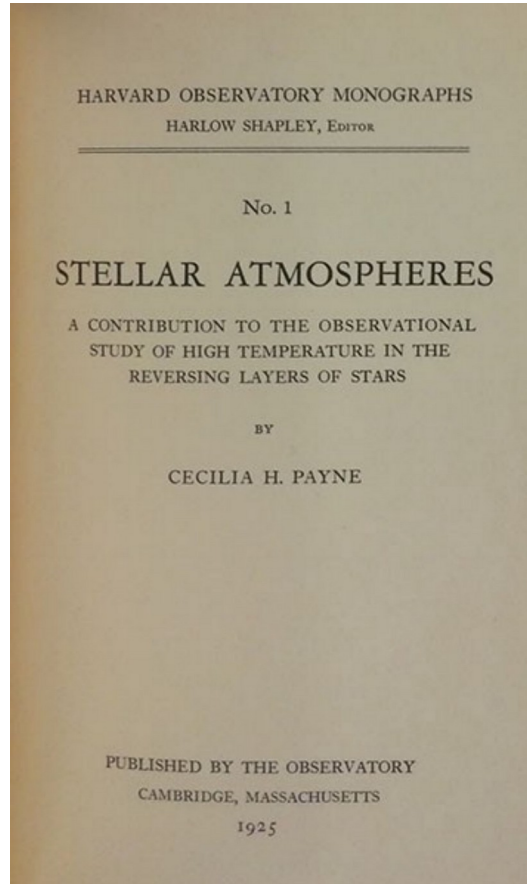


Russell (1914)



Credit: ESO

Stellar atmospheres



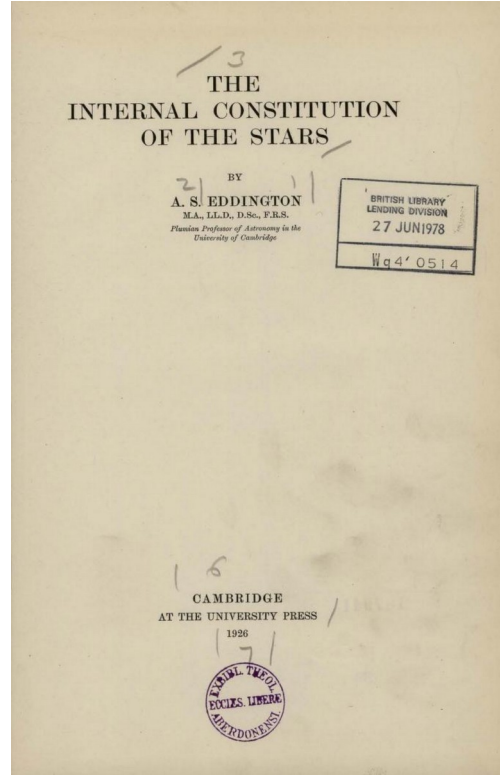
Stellar atmospheres

- Spectral classes form a sequence in effective temperature.
- Quantified the dependence between line-strengths and abundances.
- Found hydrogen to be the prominent component of stars



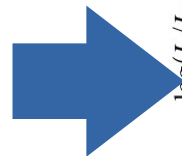
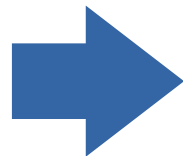
Theory of stellar structure and evolution

- ▶ Importance of radiative energy transport.
- ▶ Burning of hydrogen into helium as the energy source of the Sun (precise processes identified by von Weizsacker, Bethe and Critchfield).
- ▶ Understanding of the origin of the main sequence.

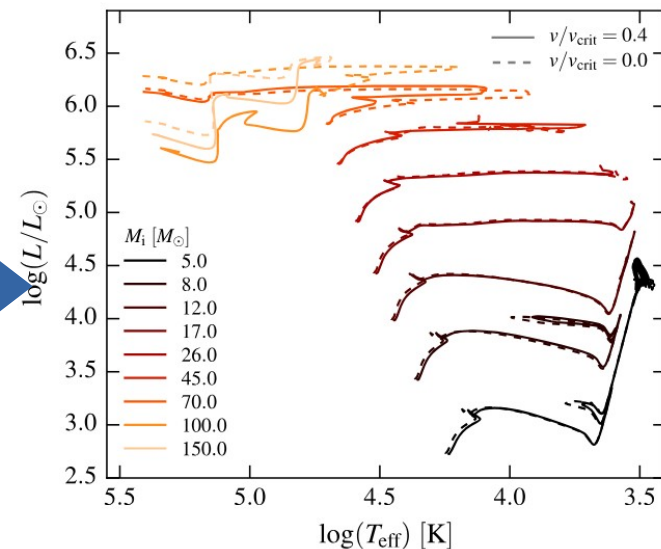


The stellar evolution black box

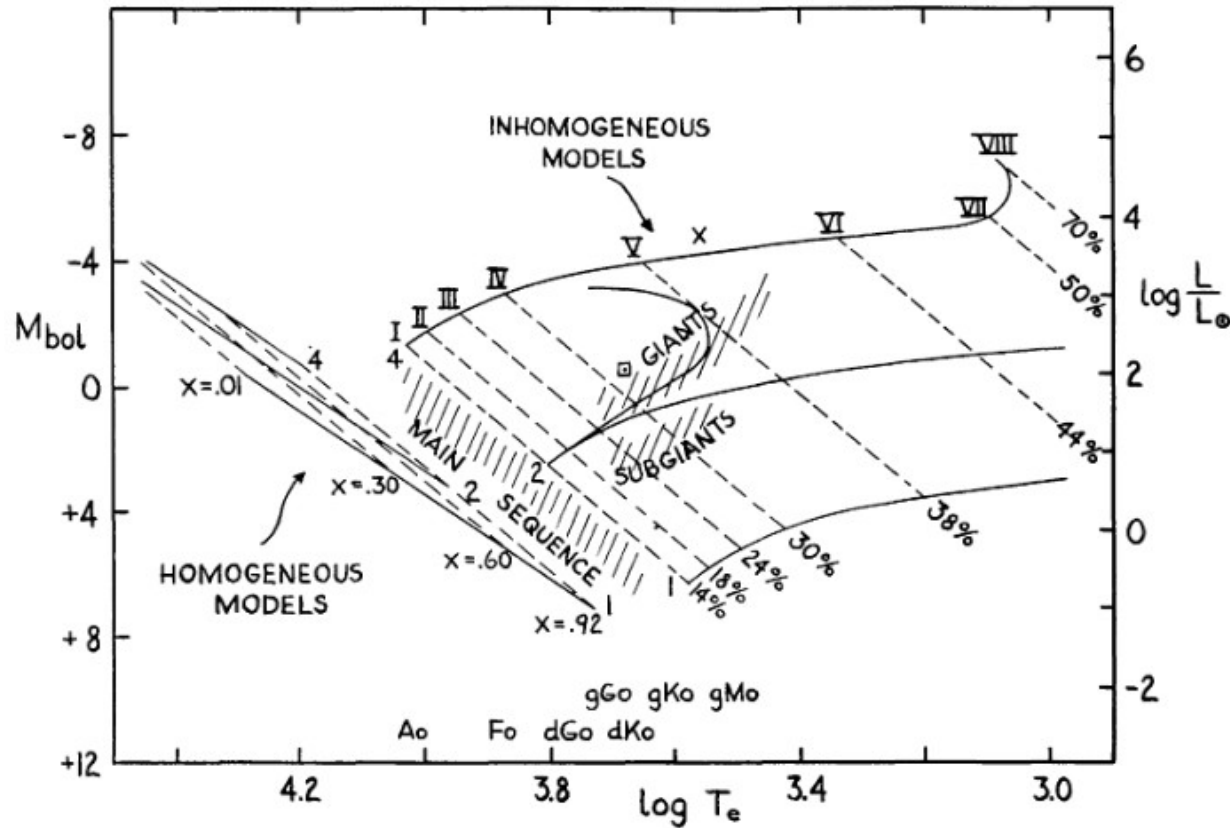
$M_{\text{initial}}, v_{\text{rot}}/v_{\text{crit}}$



Choi et al. (2016)

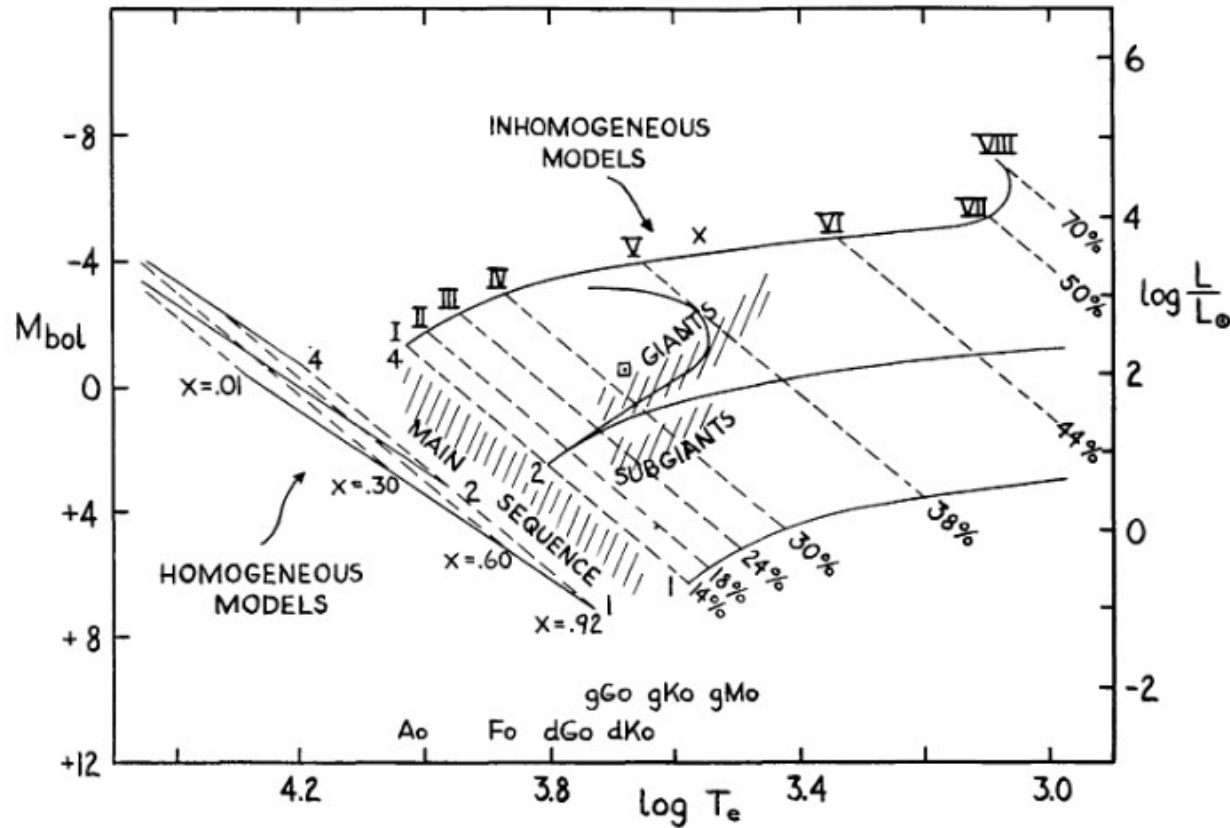


The first computer calculations



Oke & Schwarzschild (1952)
Why do stars evolve into red giants?

The first computer calculations

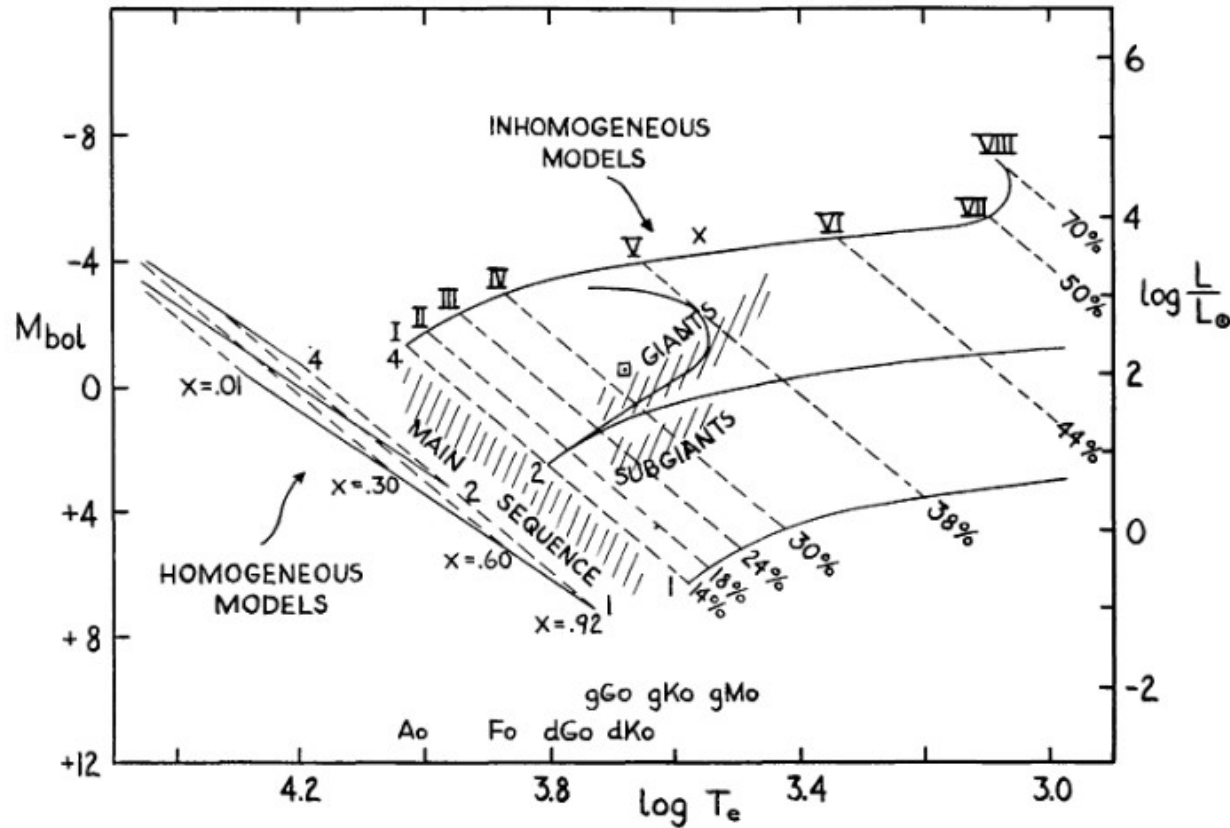


Oke & Schwarzschild (1952)
Why do stars evolve into red giants?



Desk Calculator

The first computer calculations



Oke & Schwarzschild (1952)
Why do stars evolve into red giants?

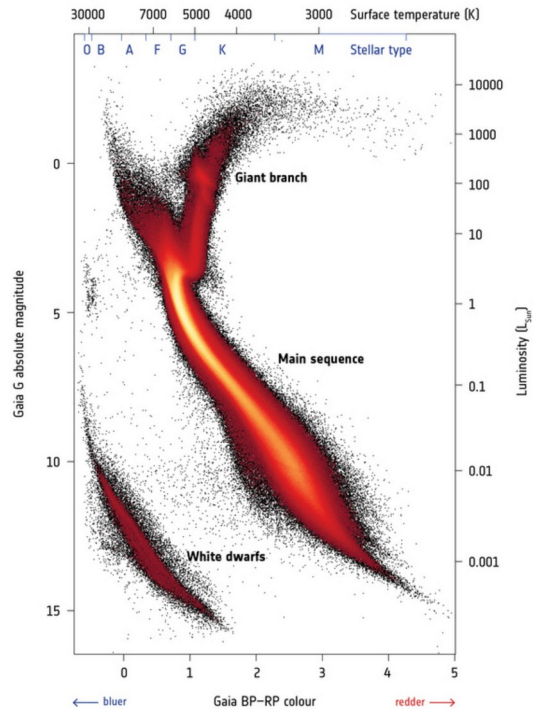


We will explore the evolution of this in future lectures

Desk Calculator

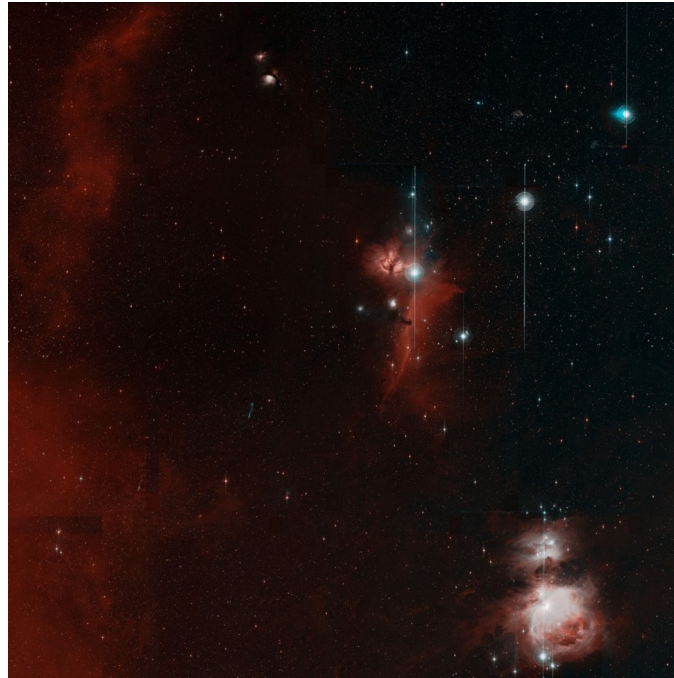
Where are we now?

Parallaxes



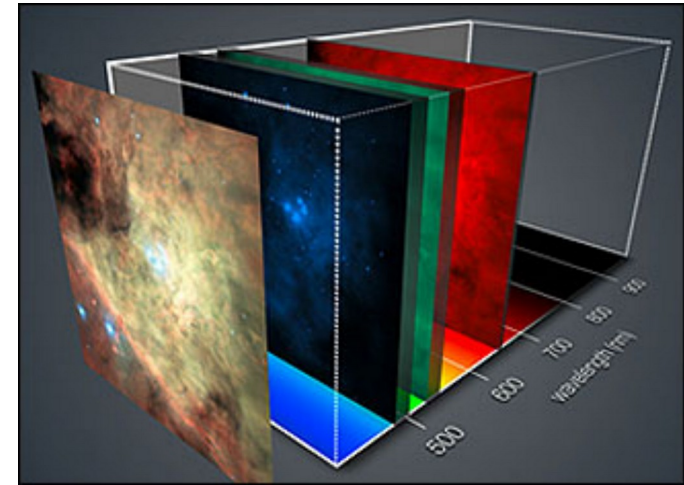
Gaia mission: parallaxes
for over a billion stars

Photometry



Zwicky Transient Facility
image of the Orion nebula

Spectroscopy



MUSE view of Orion.
Instrument at the VLT, each
pixel is a spectrum!